

ILIAD Intensive

Computational Mechanics

1. Overview & Scope

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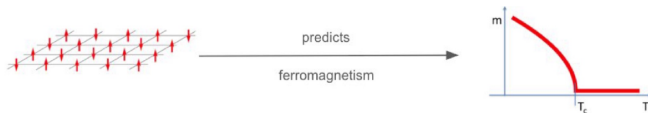
Autumn 2026

Outline

1. What is Computational Mechanics?
2. Computational Mechanics & AI Safety
3. Overview: where is the program at?
4. Scope: what will we tackle today?

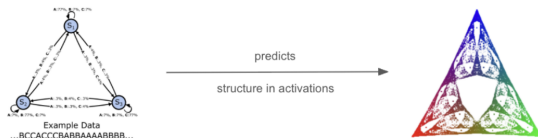
What is Computational Mechanics?

Statistical Mechanics makes predictions about the behaviour of systems by considering their **statistical** properties. E.g.



Figures from
Yuvan et al.

Computational Mechanics (comp-mech) makes predictions about the behaviour of systems by considering their **computational** properties. E.g.



Figures from
Shai et al.

Information stored by a system

Some questions that are central to comp-mech:

How much of the past does a system store?

How does the system store this information?

How does this information affect system behaviour?

We will consider each of these questions for transformers today!

Optimal prediction

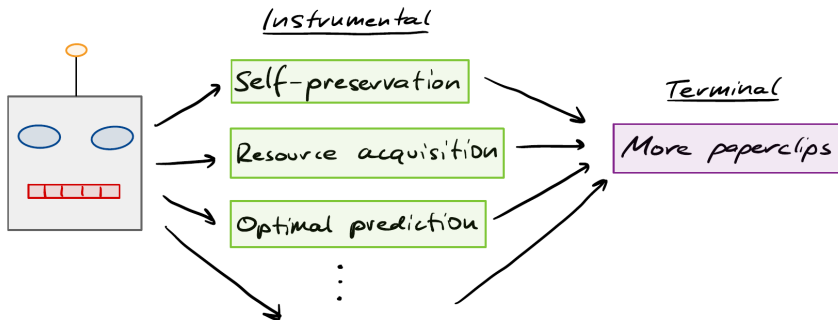
Not all systems are interesting! Comp-mech places an emphasis on those performing **optimal prediction**. In this setting the central question becomes:

What are convergent structures relevant for optimal prediction?

Insight on this question greatly constrains our search space when considering **optimal predictors**.

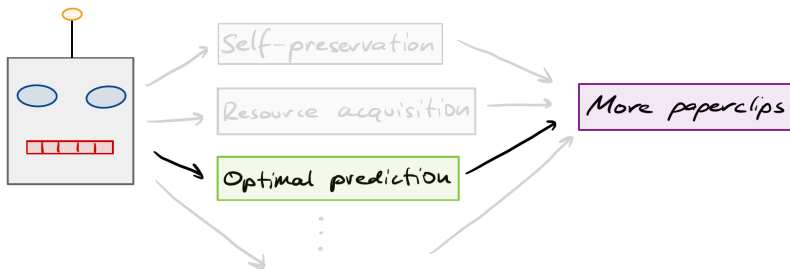
Instrumental and terminal goals

We have strong reason to believe that AI systems will pursue a range of **instrumental goals** in service of their **terminal goal**.



Optimal prediction as an intervention target

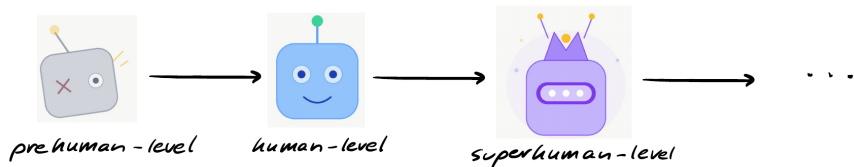
Comp-mech uses an **instrumental goal** as an intervention target point.



By considering internal representations consistent with optimal prediction the search space is reduced.

Capability scale and optimal prediction

Targeting an instrumental goal is **scale-free**: it will be relevant to current and future models.



Which contrasts **scale-sensitive** approaches e.g. AI control that are only applicable up to a fixed capability scale.

Moreover, as capabilities increase models will better approximate **optimal predictors**.

The research program

Comp-mech applied to AI safety is a young research program (~ 2 years). There is much to do!

Progress can be broken down into two orthogonal directions:

For what classes of data generating **processes** do we understand optimal prediction?

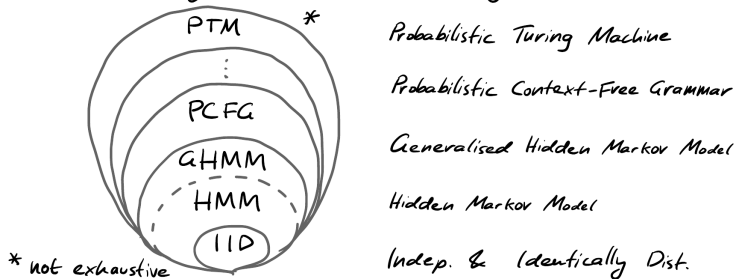
For what scale of **neural networks** have we identified the structures of optimal prediction?

Processes and generation resources

Fix a (computable) stochastic process and consider:

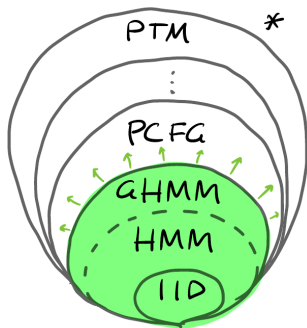
What resources are required to generate this process?

This naturally leads to a hierarchy:



Processes and optimal prediction

For what classes of data generating **processes** do we understand optimal prediction?



* not exhaustive

Probabilistic Turing Machine

Probabilistic Context-Free Grammar

Generalised Hidden Markov Model

Hidden Markov Model

Indep. & Identically Dist.

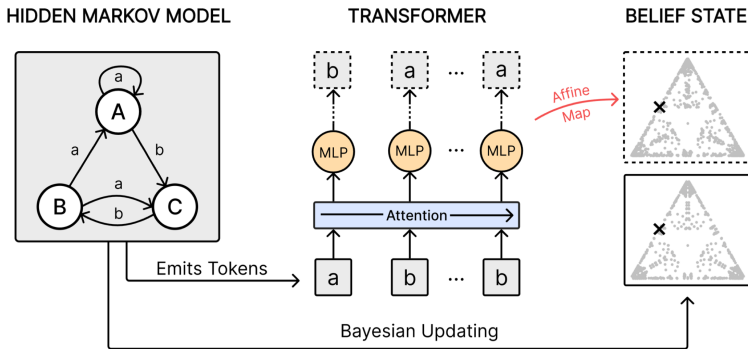
Neural networks: the scale studied so far

For what scale of **neural networks** have we identified the structures of optimal prediction?

Architecture	Parameters (approximately)
Transformer	10^5
LSTM	10^5
RNN	10^4
GRU	10^5

Very toy compared to current frontier models ($\sim 10^{12}$)!

Example: belief state geometry

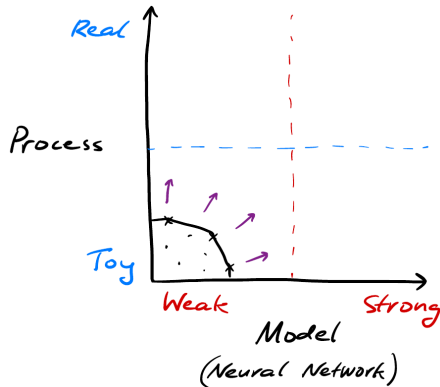


After next-token training on HMM sequences, an affine readout approximately recovers the geometry of beliefs over hidden states from residual activations.

Shai et al., *Transformers represent belief state geometry in their residual stream*. NeurIPS 2024. [arXiv:2405.15943](https://arxiv.org/abs/2405.15943).

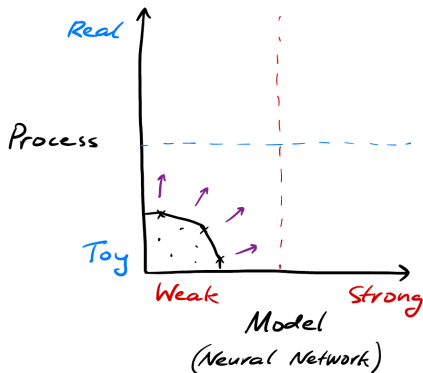
The research frontier

We are working to push the Pareto frontier:



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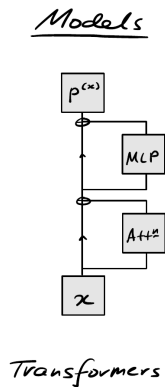
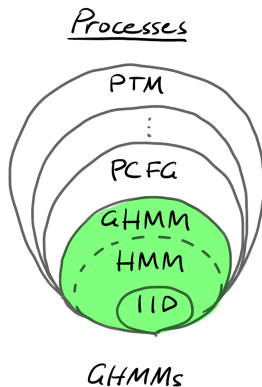


Wild Speculation

As the research program is scale-free one view is that we should develop the foundations such that future automated AI Safety researchers can pursue these ideas independently.

Scope: what will we tackle today?

We will take a slightly narrower view and consider:



Plan

- | | | |
|---|---|-------------------|
| 1 | Foundations: Hidden Markov Models | Lecture/exercises |
| 2 | Foundations: Prediction | Lecture/exercises |
| 3 | Applications: Designing Processes | Tutorial |
| 4 | Applications: Belief geometry in transformers | Reading |
| 5 | Applications: How do transformers do it? | Tutorial |
| 6 | Q & A with Adam Shai and/or Paul Riechers | |

Questions?